

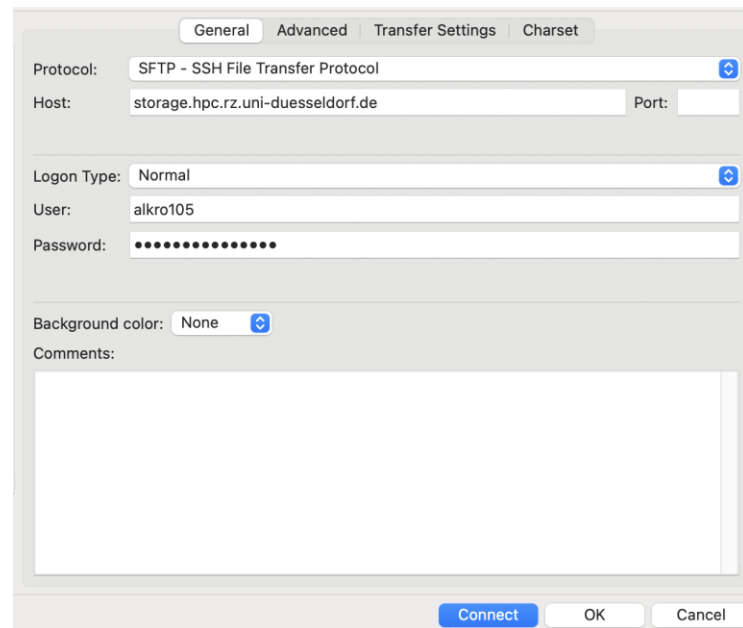
Using HPC of HHU

Requirements to log in to HPC

- You should already have an HPC account
 - It will expire with the end of this course (24.09.2025)
- You need to be in the campus network to log in to HPC. Options:
 - Connect to university wifi
 - Connected to campus network via VPN:
 - Follow instructions on <https://www.zim.hhu.de/en/servicekatalog/networks/vpn>

Managing files with FileZilla

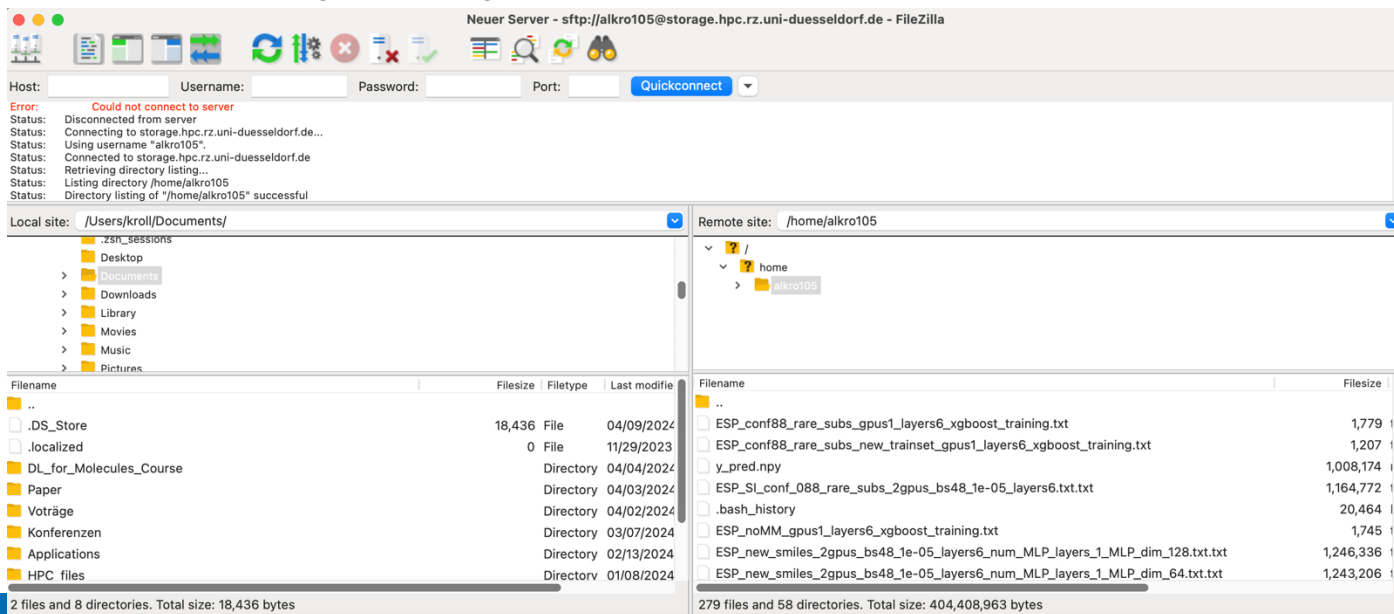
- You need to move files between your PC and the HPC
- You can do this using FileZilla: <https://filezilla-project.org/> (available for all platforms)
 - Download the FileZilla Client
- Log in to HPC using FileZilla:
 - Connect to new server (ctrl + s / cmd + s) by entering the information on the right
 - User is your university username



The screenshot shows the 'General' tab of the FileZilla client's 'Edit Site Settings' dialog. The 'Protocol' is set to 'SFTP - SSH File Transfer Protocol'. The 'Host' is 'storage.hpc.rz.uni-duesseldorf.de'. The 'Port' is empty. The 'Logon Type' is 'Normal'. The 'User' is 'alkro105'. The 'Password' field is masked with dots. The 'Background color' is set to 'None'. There is a 'Comments' text area at the bottom. At the bottom right, there are 'Connect', 'OK', and 'Cancel' buttons.

Managing files with FileZilla (2)

- After you are connected you can see
 - Left side: The files on your PC
 - Right side: Files in your HPC folder “/home/user”
- You can move files by right clicking on them



Jobs are submitted via Bash scripts

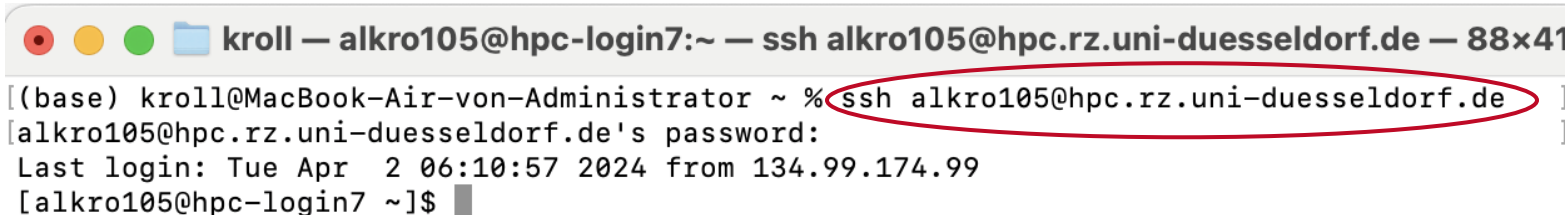
■ Example for bash script that installs the python package torch

```
1  #!/bin/bash
2  #PBS -l select=1:ncpus=1:mem=16GB ← Required resources: 1 CPU and 6GB RAM
3  #PBS -l walltime=05:00:00 ← Required runtime: After 5 hours the job will quit if not already finished
4  #PBS -A DL4MoleculesCourse ← Project name for this course. Do not change!
5  #PBS -N package_install ← Job name; choose a suitable name
6  #PBS -o "package_install.log" ← After job is finished you will find the logs in a file with this name
7
8  cd /home/alkro105/ ← Move to your home folder
9
10 module load Python/3.10.5 ← Load Python version for which you want to install the Python package
11 module load CUDA/11.7.1 ← Load CUDA to install torch with CUDA enabled
12 module load Miniconda/3
13
14 # Create a new conda environment (only need to do this once)
15 ✓ conda create -c http://conda.repo.test.hhu.de/bioconda -c http://conda.repo.test.hhu.de/main \
16 | | | -c http://conda.repo.test.hhu.de/conda-forge --override-channels --name DL4M python=3.10.5
17 |
18 # Activate the conda environment ← Command to create conda environment
19 conda activate DL4M
20
21 pip install --user -i http://pypi.repo.test.hhu.de/simple/ --trusted-host pypi.repo.test.hhu.de torch==2.0.0 ← Command to install torch 2.0.0
```

■ You need to upload this file to HPC via FileZilla if you want to execute it

Submitting jobs via terminal

- You need to connect to HPC via your terminal:



A terminal window titled "kroll — alkro105@hpc-login7:~ — ssh alkro105@hpc.rz.uni-duesseldorf.de — 88x41". The prompt is "[(base) kroll@MacBook-Air-von-Administrator ~ %]". The user enters the command "ssh alkro105@hpc.rz.uni-duesseldorf.de", which is circled in red. The terminal shows the password prompt, the last login time "Tue Apr 2 06:10:57 2024 from 134.99.174.99", and the prompt changes to "[alkro105@hpc-login7 ~]\$".

- Once logged-in, you will be in your home folder “/home/user”
 - Operating system is Linux:
 - You can see all the files in your current folder using the command `ls`
 - You can submit a job using the command `qsub name_of_bash_script.sh`
 - You can see the status of your jobs
 - `qstat -u username`
 - Status “Q” stands for queued; “R” stands for running; After job is executed it will vanish from this list
 - You can also monitor your jobs on myjam3.hhu.de

Running a Python Script

- Implement Bash script to execute Python script:

```
1  #!/bin/bash
2  #PBS -l select=1:ncpus=1:ngpus=1:mem=16GB
3  #PBS -l walltime=40:00:00
4  #PBS -A DL4MoleculesCourse
5  #PBS -N package_install
6  #PBS -o "package_install.log"
7
8  cd /home/alkro105/
9
10 module load Python/3.10.5
11 module load CUDA/11.7.1 #only if you need CUDA (for running torch on GPU)
12 module load Miniconda/3
13
14
15 # Activate the conda environment
16 conda activate DL4ML
17
18 #Run python script
19 python3 /home/alkro105/python_script.py
```

Only set ngpus if you need a GPU!

- Upload your Python script and your bash script to your HPC folder via FileZilla
- Submit Bash script via terminal

- For quick Debugging, you can use the Jupyterhub: <https://jupyter.hpc.rz.uni-duesseldorf.de/>

Server Options

Project
DL4mol ▼

Cores
1 ▼

Memory
50 GB ▼

GPUs
1 ▼

Walltime
4 Hours ▼

Queue
default ▼

Start

Jupyterhub (2)

- You can run Jupyter Notebooks that you uploaded to your home folder
- Notebooks will run in a default environment with most important packages installed
 - Follow these instructions to change this environment:
 - <https://wiki.hhu.de/display/HPC/Jupyter>
- GPU will be the NVIDIA GTX1080ti